TANGENT SPACES OF F ARE FINITE DIMENSIONAL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 6211-6215 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 6217-6234, the complete author proof (16 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 40 of queue_artin.json (pb=799, stmt 6211-6215, proof 6217-6234).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07X1: sources/stacks-artin/pinned-original-source.tex, source lines 1030–1037. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GI6: sources/stacks-artin/pinned-original-source.tex, source lines 6175–6178. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GH8: sources/stacks-artin/pinned-original-source.tex, source lines 5138–5153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Situation 0GH8 the tangent spaces of the functor F are finite dimensional.*

Proof. In the proof of Lemma 0GI6 we have seen that $F(V) = U'(V) \amalg W(V)$ if V is the spectrum of an Artinian local ring. The tangent spaces are computed entirely from evaluations of F on such schemes over S. Hence it suffices to prove that the tangent spaces of the functors U' and W are finite dimensional. For U' this follows from Lemma 07X1. Write $W = \operatorname{colim} W_n$ as in the proof of Lemma 0GI6. Then

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we see that the tangent spaces of W are equal to the tangent spaces of W_2, as to get at the tangent space we only need to evaluate W on spectra of Artinian local rings $(A, \mathfrak{m})$ with $\mathfrak{m}^2 = 0$. Then again we see that the tangent spaces of W_2 have finite dimension by Lemma 07X1. $\square$